

**BANGLADESH UNIVERSITY OF ENGINEERING AND
TECHNOLOGY**



Department of Electrical and Electronic Engineering

Course No. : EEE 212

Course Title: Numerical Technique Laboratory

Project Report

Project Name:

**“Solving Steady State Heat Conduction Equation with Heat
Generation In 2-D Space Using Numerical Method”**

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Introduction:

Heat transfer is a ubiquitous phenomenon that occurs in nature and technology, and one of the most important mechanisms of heat transfer is conduction. Conduction involves the transfer of heat through a material by molecular interactions, and it plays a crucial role in a wide range of practical applications. In this context, we will discuss the concept of “Steady state heat conduction” and solve the equations in order to plot the values using MATLAB so that we can observe and realize how the heat is being conducted through the material.

“Steady-state heat conduction” refers to a state in which the temperature gradient within a material is constant over time, meaning that there is no net heat flow. In other words, the rate of heat transfer into a material is equal to the rate of heat transfer out of the material. Understanding the mechanisms and principles underlying the conduction of steady-state heat transfer is crucial for optimizing the performance and efficiency of many engineering processes. By controlling the rate of heat transfer, we can minimize energy waste, reduce costs, and improve the overall sustainability of our activities.

In the following project, we have come up with a solution for the steady state heat conduction equation by using the numerical method in 2-D space. we will explore the basic principles, equations, and applications of steady-state heat conduction in greater detail. We will see how it is affected by various factors such as the thermal conductivity of the material, the temperature difference, and the cross-sectional area of the material. We will also discuss how these factors can be optimized to improve the efficiency and effectiveness of heat transfer in various practical applications. Steady-state heat conduction is a crucial concept that underlies many aspects of heat transfer and has numerous practical applications in science and engineering. Through this discussion, we hope to deepen our understanding of this topic and its significance and applications in various fields.

Theory:

The steady-state heat conduction equation describes the transfer of heat through a material in a state of constant temperature. In many practical applications, there is heat generation within the material, which complicates the equation and makes it difficult to solve analytically.

In such cases, numerical methods are often used to solve the equation and predict the temperature distribution within the material. In conduction steady-state heat transfer, the temperature gradient is established within a solid material, where heat is transferred from a hotter region to a colder region.

The Fourier's law of heat conduction in a material is given by:

$$q = -kA(dT/dx)$$

where q is the rate of heat transfer per unit time, k is the thermal conductivity of the material, A is the cross-sectional area of the material, T is the temperature, and x is the distance along the material. This equation shows that the rate of heat transfer is directly proportional to the thermal conductivity of the material and the cross-sectional area, and inversely proportional to the temperature gradient and the distance along the material.

The differential equation of the steady state heat conduction is:

$$(\delta^2 T / \delta x^2) + (\delta^2 T / \delta y^2) + \frac{g}{k} = 0$$

One standard numerical method used to solve the steady state heat conduction equation with heat generation in 2-D space is the finite difference method. This method involves dividing the material into a grid of nodes and approximating the temperature at each node using a finite difference equation. The finite difference equation relates the temperature at a given node to the temperatures at adjacent nodes and the heat generation rate at that node.

The finite difference equation can be derived from the steady state heat conduction equation using Taylor series expansions and truncating higher-order terms. Once the finite difference equation is derived, it can be solved iteratively using a numerical algorithm such as the Gauss-Seidel method or the Jacobi method. The solution obtained using the finite difference method gives the temperature distribution within the material in a steady state condition. This

information can be used to optimize the design of various heat transfer systems and processes.

Procedure & Equation Solving:

1. Formulation of Governing Differential Equation:
- Energy Balance (Law of Conservation of Energy):

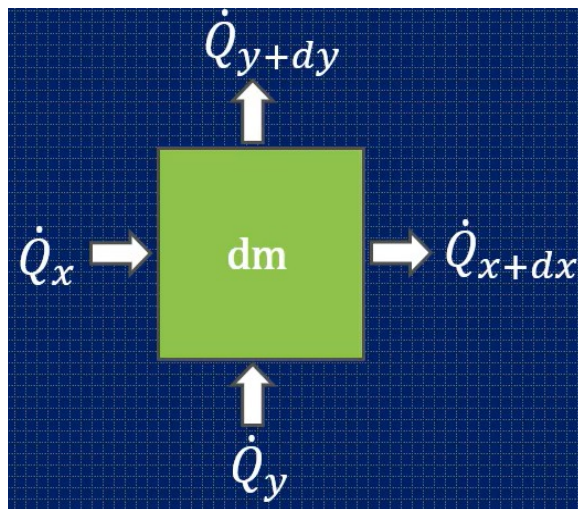


Fig: A Small heat conductor object of mass dm

The Energy balance equation is:

$$(\text{Rate of Energy})_{\text{in}} + (\text{Rate of Energy})_{\text{out}} + (\text{Rate of Energy})_{\text{generated}} = (\text{Rate of Energy})_{\text{Change}}$$

$$\text{Or, } (\dot{Q}_x + \dot{Q}_y) - (\dot{Q}_{x+dx} + \dot{Q}_{y+dy}) + \dot{Q}_{\text{gen}} = dm \cdot C_p \frac{dT}{dt}$$

$$\text{Hence, } \dot{Q}_{\text{gen}} = g = P / (4 \cdot \pi \cdot ((x-a)^2 + (y-b)^2)); \quad [p = \text{power of the heat source}]$$

a = abscissa of the position of the heat source

b = ordinate of the position of the heat source

$$\text{Or, } (\dot{Q}_x + \dot{Q}_y) - (\dot{Q}_{x+dx} + \dot{Q}_{y+dy}) + g = dm \cdot C_p \frac{dT}{dt}$$

For steady state operation, $\frac{dT}{dt} = 0$

$$\text{Or, } (\dot{Q}_x + \dot{Q}_y) - (\dot{Q}_{x+dx} + \dot{Q}_{y+dy}) + g = 0 \dots (1)$$

● Taylor Series Expansion:

We know,

$$f(x+dx) = f(x) + f'(x)\frac{dx}{1!} + f''(x)\frac{dx^2}{2!} + f'''(x)\frac{dx^3}{3!} + \dots \text{ [From Taylor Series]}$$

$$\text{or, } \dot{Q}_{x+dx} = \dot{Q}_x + \dot{Q}_x' \frac{dx}{1!} + \dot{Q}_x'' \frac{dx^2}{2!} + \dot{Q}_x''' \frac{dx^3}{3!} + \dots$$

Neglecting the higher order terms,

$$\dot{Q}_{x+dx} = \dot{Q}_x + \dot{Q}_x' \frac{dx}{1!}$$

Similarly in the Y-direction,

$$\dot{Q}_{y+dy} = \dot{Q}_y + \dot{Q}_y' \frac{dy}{1!}$$

Replacing the values of $\dot{Q}_{x+dx}$ & $\dot{Q}_{y+dy}$ in eq_n (1),

$$(\dot{Q}_x + \dot{Q}_y) + (\dot{Q}_x + \dot{Q}_x' \frac{dx}{1!} + \dot{Q}_y + \dot{Q}_y' \frac{dy}{1!}) + g = 0$$

$$\text{Or, } -\dot{Q}_x' \frac{dx}{1!} - \dot{Q}_y' \frac{dy}{1!} + g = 0$$

$$\text{Or, } \dot{Q}_x' \frac{dx}{1!} + \dot{Q}_y' \frac{dy}{1!} + g = 0 \dots (2)$$

● Fourier's law of heat conduction:

Heat conduction in the X-direction, $\dot{Q}_x = -kA \frac{dT}{dx}$

Heat conduction in the Y-direction, $\dot{Q}_y = -kKA \frac{dT}{dy}$

Here,

K = Thermal Conductivity (W/mK)

A = Area perpendicular to heat flow (m²)

Now,

From eqⁿ (2),

$$\dot{Q}_x \frac{dx}{1} + \dot{Q}_y \frac{dy}{1} + g = 0$$

$$\text{Or, } \frac{d}{dx}(\dot{Q}_x)dx + \frac{d}{dy}(\dot{Q}_y)dy + g = 0$$

Replacing the values of $\dot{Q}_x$ & $\dot{Q}_y$,

$$\text{Or, } \frac{d}{dx}(-kKA \frac{dT}{dx})dx + \frac{d}{dy}(-kKA \frac{dT}{dy})dy + g = 0$$

$$\text{Or, } \frac{d}{dx}(-kKdy \frac{dT}{dx})dx + \frac{d}{dy}(-kKdx \frac{dT}{dy})dy + g = 0$$

$$\text{Or, } \frac{d}{dx}(-kK \frac{dT}{dx})dx dy + \frac{d}{dy}(-kK \frac{dT}{dy})dx dy + g = 0$$

$$\text{Or, } \frac{d}{dx}(K \frac{dT}{dx}) + \frac{d}{dy}(K \frac{dT}{dy}) + g = 0 \dots \dots \dots (3)$$

Here, K is a constant.

Then,

$$K \frac{d}{dx}(\frac{dT}{dx}) + K \frac{d}{dy}(\frac{dT}{dy}) + g = 0$$

$$\text{Or, } \frac{d}{dx}(\frac{dT}{dx}) + \frac{d}{dy}(\frac{dT}{dy}) + g = 0$$

$$\therefore (d^2T/dx^2) + (d^2T/dy^2) + g/K = 0 \dots \dots \dots (4)$$

This is the differential equation for 2D steady state heat conduction.

2. Discretization of Domain into Subdomain:

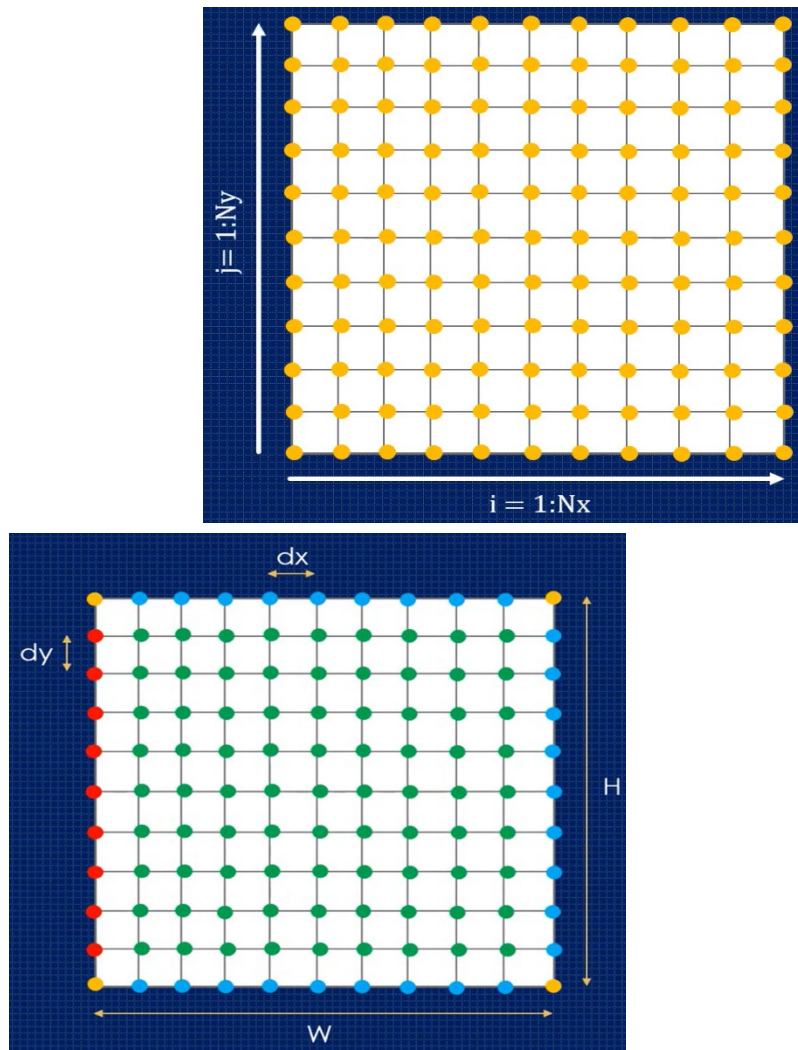


Fig.: Discretization

In the figure, we can see that the domain is discretized into many sub-domain. There are grid points that divided the width and height domain into many grids or sub-domains.

Here,

N_x = Number of Grid-points in the x-axis

N_y = Number of Grid-points in the y-axis

W = Width of the object

H = Height of the object

And,

$$dx = W / (N_x - 1)$$

$$dy = H / (N_y - 1)$$

3. Converting Differential Equations into Algebraic Equations:

From Taylor's Series Expansion,

$$f(x+dx) = f(x) + f'(x)\frac{dx}{1!} + f''(x)\frac{dx^2}{2!} \dots \dots \dots (i) \text{ [Neglecting higher order terms]}$$

$$f(x-dx) = f(x) - f'(x)\frac{dx}{1!} + f''(x)\frac{dx^2}{2!} \dots \dots \dots (ii) \text{ [Same as before]}$$

Adding (i) & (ii),

$$f(x+dx) + f(x-dx) = 2f(x) + 2f''(x)\frac{dx^2}{2!}$$

$$\text{Or, } f(x+dx) + f(x-dx) = 2f(x) + f''(x)dx^2$$

$$\therefore f''(x) = (f(x+dx) + f(x-dx) - 2f(x)) / dx^2 \dots \dots \dots (iii)$$

$$\therefore (d^2T/dx^2) = (T(x+dx, y) + T(x-dx, y) - 2T(x, y)) / dx^2$$

$$\therefore (d^2T/dy^2) = (T(x, y+dy) + T(x, y-dy) - 2T(x, y)) / dy^2$$

From the main eq_n,

$$(d^2T/dx^2) + (d^2T/dy^2) + g/K = 0$$

$$\text{Or, } (T(x+dx, y) + T(x-dx, y) - 2T(x, y)) / dx^2 + (T(x, y+dy) + T(x, y-dy) - 2T(x, y)) / dy^2 + g/K = 0$$

$$\text{Or, } T(x,y) = ((T(x+dx, y) + T(x-dx, y))/dx^2 + (T(x, y+dy) + T(x, y-dy))/dy^2)/$$

$$(2/dx^2 + 2/dy^2) + g \cdot dx \cdot dy / K$$

Assuming $dx = dy$,

$$T(x,y) = (T(x+dx, y) + T(x-dx, y) + T(x, y+dy) + T(x, y-dy))/4 + g \cdot dx \cdot dy / 4K$$

$$\therefore T(i,j) = (T(i+1, j) + T(i-1, j) + T(i, j+1) + T(i, j-1))/4 + g \cdot dx \cdot dy / 4K$$

This is the algebraic equation of the corresponding differential equation. And now, we will use MATLAB to solve this equation.

MATLAB Code:

-

```
classdef heateqnsolver < matlab.apps.AppBase
```

```
    methods (Access = private)
```

```
        function D3plotgraph(app,h,W,H,T)
```

```
            x = 0:h:W;
```

```
            y = 0:h:H;
```

```
            c = colorbar(app.UIAxes)
```

```
            colormap(app.UIAxes, jet)
```

```
            surf(app.UIAxes, x,y,T');

```

```
        end
```

```
        function D2plotgraph(app,h,W,H,T)
```

```
            x = 0:h:W;
```

```
            y = 0:h:H;
```

```
            c = colorbar(app.UIAxes)
```

```
            colormap(app.UIAxes, jet)
```

```
            contourf(app.UIAxes,x,y,T');

```

```

end

end

% Callbacks that handle component events

methods (Access = private)

    % Value changed function: TextArea

    function TextAreaValueChanged(app, event)

    end

    % Button pushed function: NextButton

    function NextButtonPushed(app, event)

        app.TabGroup.SelectedTab = app.ParametersTab;

    end

    % Button pushed function: NextButton_2

    function NextButton_2Pushed(app, event)

        app.TabGroup.SelectedTab = app.PlotTab;

    end

    % Button pushed function: NextButton_3

    function NextButton_3Pushed(app, event)

        app.TabGroup.SelectedTab = app.DevelopersTab;

    end

    % Callback function

    function NextButton_4Pushed(app, event)

    end

    % Button pushed function: PlotButton

    function PlotButtonPushed(app, event)

        %% Geometry of Domain

        W = app.WidthoftheRectangularplateEditField.Value;

```

```
H = app.HeightoftheRectangularplateEditField.Value;
```

```
Nx = 51;
```

```
h = min(W,H)/(Nx-1);
```

```
Nx = round(W/h) + 1;
```

```
Ny = round(H/h) + 1 ;
```

```
W = (Nx -1)*h;
```

```
H = (Ny -1)*h;
```

```
a = app.xEditField.Value;
```

```
b = app.yEditField.Value;
```

```
P = app.PoweroftheheatsourceWEditField.Value;
```

```
q = @(x,y)((x-a).^2+(y-b).^2);
```

```
%% Boundary and Initial Conditons
```

```
T = 25*ones(Nx,Ny);
```

```
TL = app.TemperatureontheLeftSideEditField.Value;
```

```
TR = app.TemperatureontheRightSideEditField.Value;
```

```
TT = app.TemperatureontheUpperSideEditField.Value;
```

```
TB = app.TemperatureontheLowerSideEditField.Value;
```

```
T(1,2:Ny-1) = TL;
```

```
T(2:Nx-1,Ny) = TT;
```

```
T(Nx,2:Ny-1) = TR;
```

```
T(2:Nx-1,1) = TB;
```

```
T(1,1) = (TL+TB)/2;
```

```
T(1,Ny) = (TL+TT)/2;
```

```
T(Nx,Ny) = (TR+TT)/2;
```

```
T(Nx,1) = (TR+TB)/2;
```

```

Iter = 0;

k =app.ThermalConductivityEditField.Value;

Epsilon = 1e-5;

Error = 50;

x = 0:h:W;

y = 0:h:H;

while (Error>Epsilon)

    Iter = Iter+1;

    Told = T;

    for j = 2 : Ny-1

        for i = 2:Nx-1

            g = q(x(i), y(j));

            if g == 0

                g = (q(x(i-1), y(j)) + q(x(i), y(j-1)) + q(x(i+1), y(j)) + q(x(i), y(j+1)))) / 4;

            end

            g = P / (4*pi*g);

            T(i,j) = (T(i+1,j)+T(i-1,j)+T(i,j+1)+T(i,j-1))/4 + g/(4*k)*h^2 ;

        end

    end

    Error = sqrt(sum(sum((T-Told).^2)));

end

%% Plotting the results

value = app.ChooseoneoptionDropDown.Value

if(strcmp(value,'2-D Plot')==1)

    D2plotgraph(app,h,W,H,T);

    app.EditField_3.Value = 'Temparature Distribution.';

```

```

elseif(strcmp(value,'3-D Plot')==1)

    D3plotgraph(app,h,W,H,T);

    app.EditField_3.Value = '3-D Plot of T';

end

end

% Value changed function: xEditField

function xEditFieldValueChanged(app, event)

end

% Value changed function: yEditField

function yEditFieldValueChanged(app, event)

end

% Callback function

function PlotButton_2Pushed(app, event)

end

% Button pushed function: ExitButton

function ExitButtonPushed(app, event)

    % Ask the user if they want to exit the program

    choice = questdlg('Are you sure you want to exit the program?', ...

        'Exit Confirmation', ...

        'Yes','No','No');

    % If the user clicks "Yes", close the app window and exit the program

    if strcmp(choice, 'Yes')

        delete(app.UIFigure);

    end

end

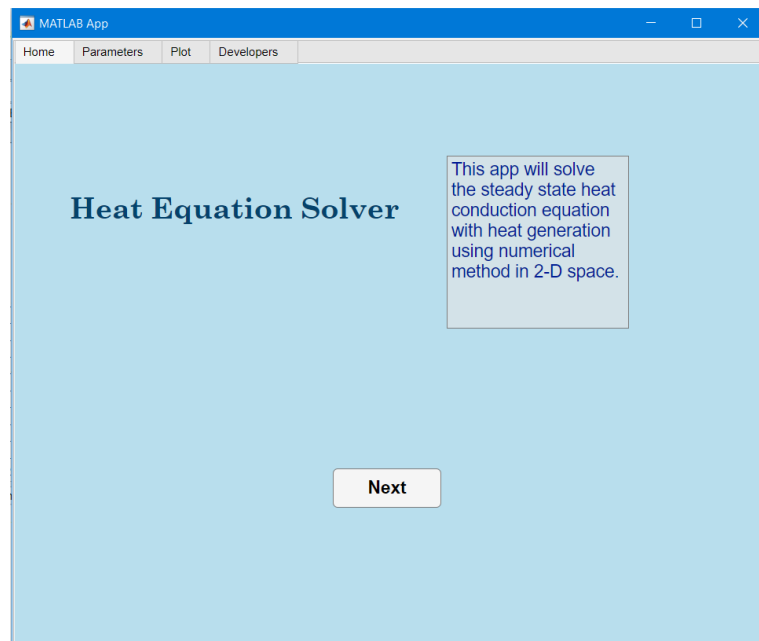
end
end

```

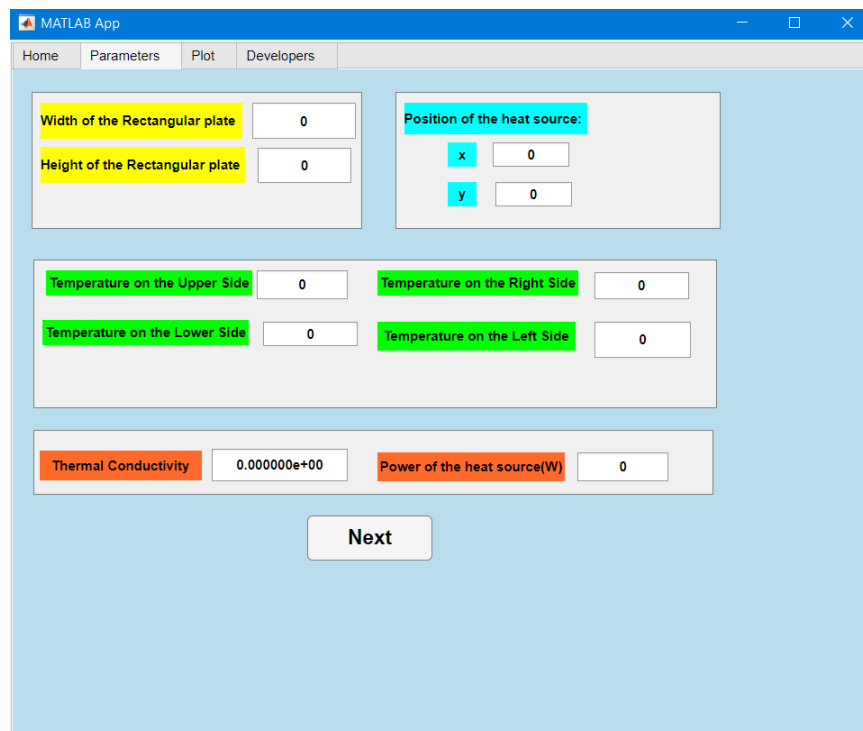
THE GRAPHICAL USER INTERFACE

At this part of the program, we created a graphical user interface for the user to use the program easily. The main program starts with this part of the project. We'll describe the step-by-step procedure below to understand better.

This is the homepage of the application called “heateqnsolver”.

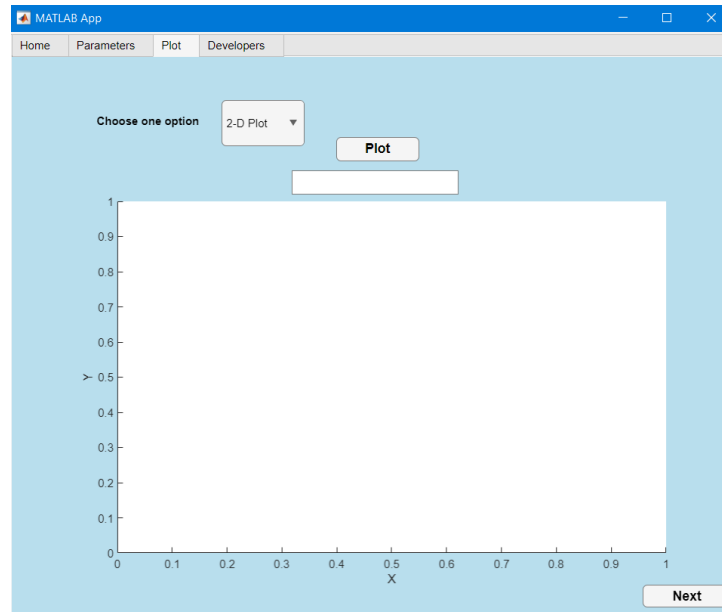


On this tab, the user will insert the value of the parameters.

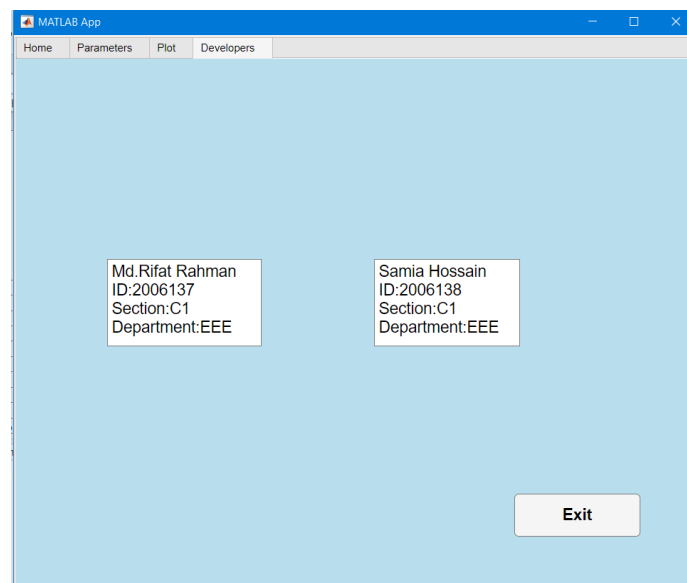


In this tab, the user will plot the graph. There will be 2 options: 2-D graph and 3-D graph.

The user can choose any of them and the plot will be shown.

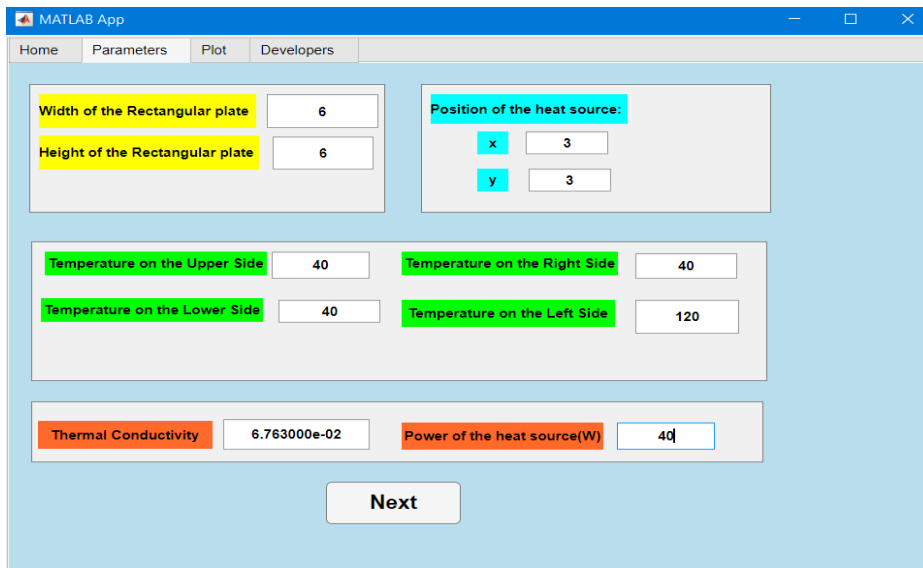


This is the last tab. The user can exit the program by clicking on exit.



Sample Output:

The following sample input was taken to run the application.

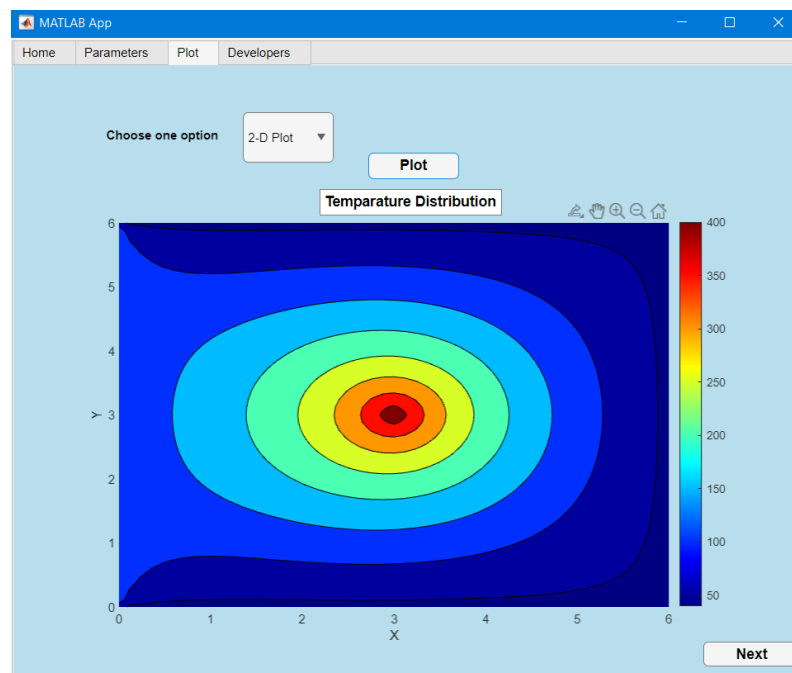


The MATLAB App interface displays the following input parameters:

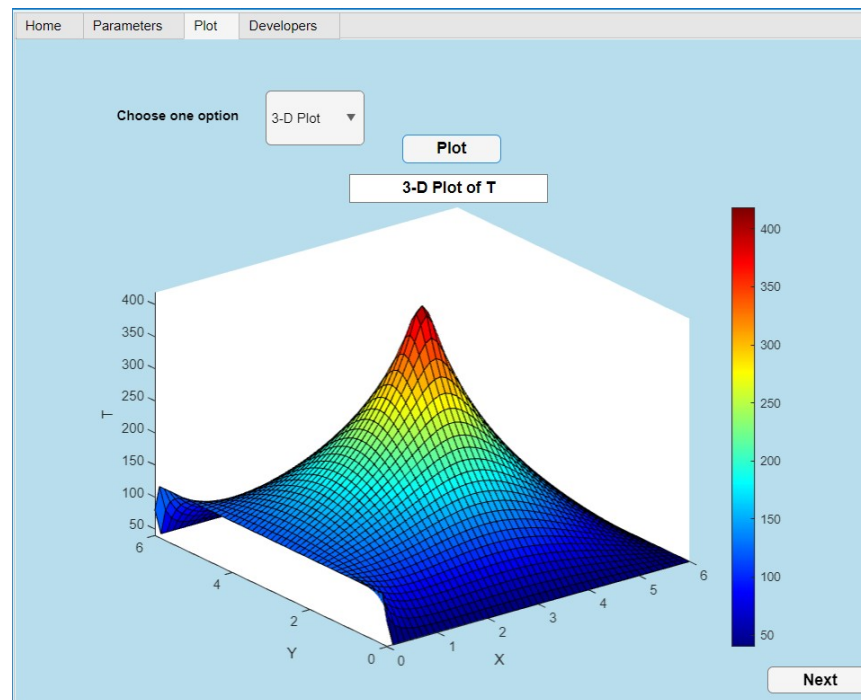
- Width of the Rectangular plate:** 6
- Height of the Rectangular plate:** 6
- Position of the heat source:**
 - x:** 3
 - y:** 3
- Temperature on the Upper Side:** 40
- Temperature on the Right Side:** 40
- Temperature on the Lower Side:** 40
- Temperature on the Left Side:** 120
- Thermal Conductivity:** $6.763000e-02$
- Power of the heat source(W):** 40

A **Next** button is located at the bottom center of the parameter input section.

The following 2-D graph was generated.



The following 3-D graph was generated.



Application:

Steady-state heat conduction has a wide range of practical applications in various fields, including engineering, physics, and materials science. Here are some examples:

1. **Thermal insulation:** Steady-state heat conduction is used to design and optimize thermal insulation materials for buildings, refrigerators, and other appliances. By understanding the steady-state heat flow through materials, engineers can select and optimize materials to minimize heat transfer and increase energy efficiency.
2. **Heat exchangers:** Steady-state heat conduction is used to design and optimize heat exchangers in various applications, such as cooling systems in automobiles, power plants, and air conditioning systems. By understanding the steady-state heat flow through different materials and fluids, engineers can design efficient heat exchangers that transfer heat from one medium to another.

3. Electronic cooling: Steady-state heat conduction is used in the design of electronic devices and cooling systems. By understanding the steady-state heat flow through electronic components, engineers can design efficient cooling systems to prevent overheating and improve performance.
4. Geothermal energy: Steady-state heat conduction is used to design and optimize geothermal energy systems. By understanding the steady-state heat flow through the earth's crust, engineers can design efficient systems to extract heat from the earth's core.

Overall, steady-state heat conduction and solving its equation is a critical tool for engineers and scientists to understand heat flow and optimize the design and performance of various systems and materials.

Conclusion:

Steady-state heat conduction plays an important role in the field of thermodynamics, and it has many numerous practical applications. One of the most significant benefits of steady-state heat conduction is its potential for energy efficiency. By understanding how heat is transferred within a system, we can design and operate systems that are more energy-efficient, leading to cost savings and reduced environmental impact. On the other hand, temperature control is being done by steady-state heat conduction, which is crucial for maintaining a comfortable indoor environment in buildings and vehicles. Effective temperature control also ensures the safe and efficient operation of many systems. Additionally, steady-state heat conduction plays a vital role in promoting sustainability. By designing and operating systems that effectively manage heat transfer, we can reduce energy consumption and minimize environmental impact. Steady-state heat conduction is essential for ensuring the safe operation of many systems. By designing systems that effectively manage heat transfer, we can prevent equipment failures, fires, and other safety hazards. all, the concept and application of steady-state heat conduction play an important role in every sphere of our life. By understanding this concept and numerically solving its equations, we can design and operate systems that are more energy-efficient, safe, and sustainable, while also providing comfortable indoor environments for occupants and making life easier.